

Literature Review: DLM Architectures/Quantization

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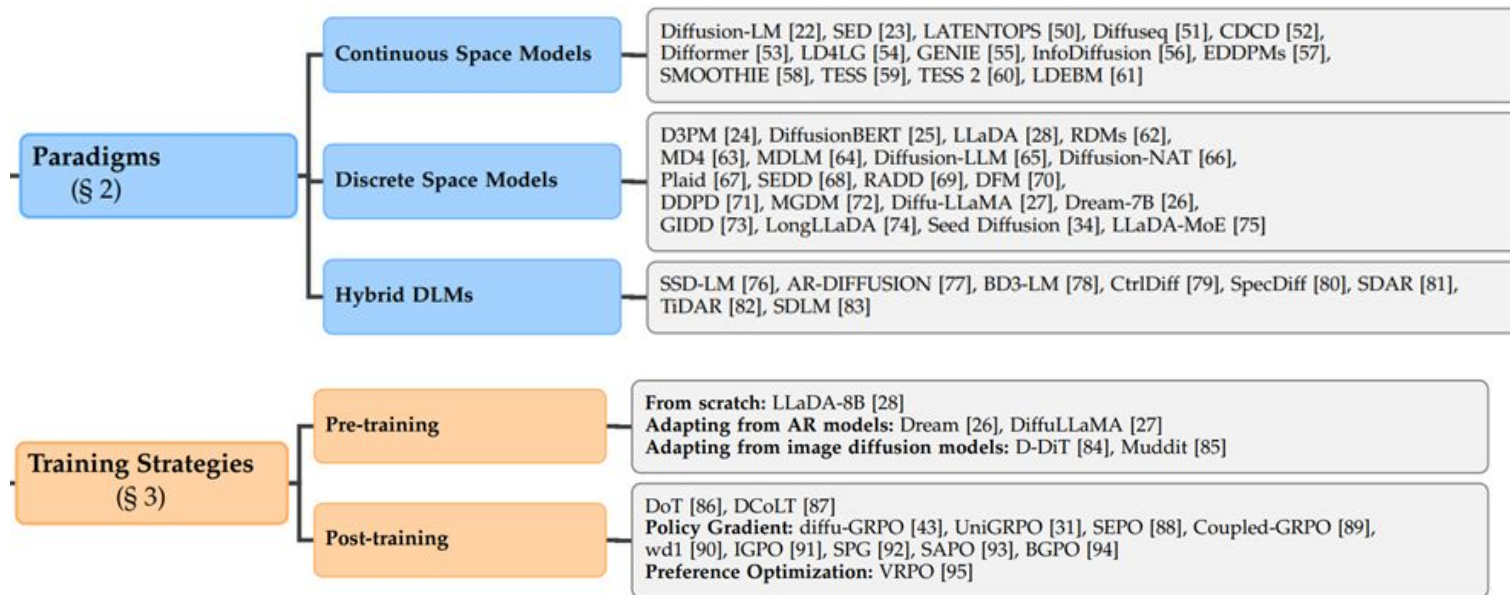
Outline

- Classification of Diffusion Language Models (DLMs)
- Overview of Best Performing DLM Architectures
- Quantization of DLM

Why use DLMs?

- Parallel generation of tokens
- Bidirectional context
- Iterative refinement of output
- Controllability
- Adjustable inference process through diffusion steps to balance speed and output

Classification Of DLMs



Discrete/Hybrid Models

- D3PM
- Masked Diffusion Language Model (MDLM)
- Block Diffusion Language Model (BDLM)

J. Austin, D. D. Johnson, J. Ho, D. Tarlow, and R. van den Berg, “Structured Denoising Diffusion Models in Discrete State-Spaces,” Feb. 22, 2023, *arXiv*: arXiv:2107.03006.

M. Arriola *et al.*, “Block Diffusion: Interpolating Between Autoregressive and Diffusion Language Models,” May 17, 2025, *arXiv*: arXiv:2503.09573.

LLaDA-8B Model

- MDLM trained on 2.3 trillion tokens
- Trained on multilingual data, math data and high quality code
- LLaDA architecture:
 - Forward data masking process
 - Trained mask predictor for the reverse process
- LLaDa minimizes an upper bound on the negative log likelihood of data distribution

$$\mathcal{L}(\theta) \triangleq -\mathbb{E}_{t,x_0,x_t} \left[\frac{1}{t} \sum_{i=1}^L \mathbf{1}[x_t^i = \mathbf{M}] \log p_{\theta}(x_0^i | x_t) \right],$$

LLaDA-8B Architecture

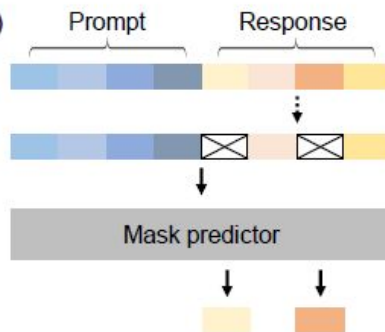
(a) Mask all tokens independently



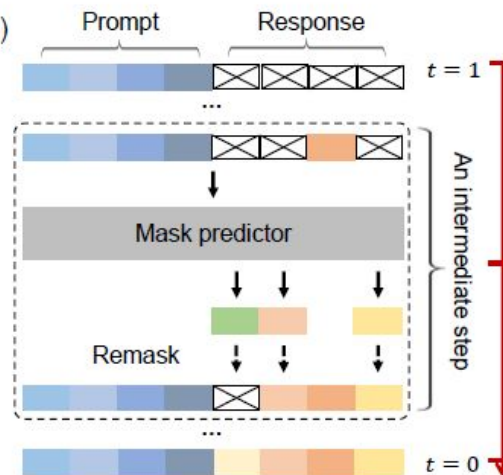
Mask predictor



(b)



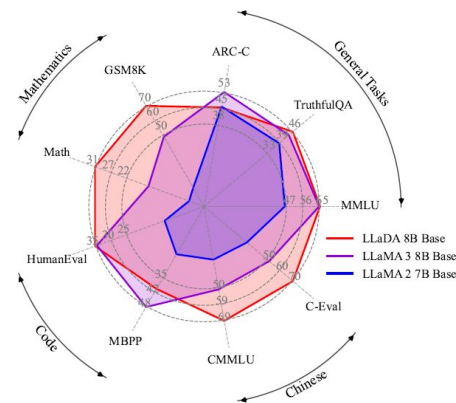
(c)



LLaDA-8B Results

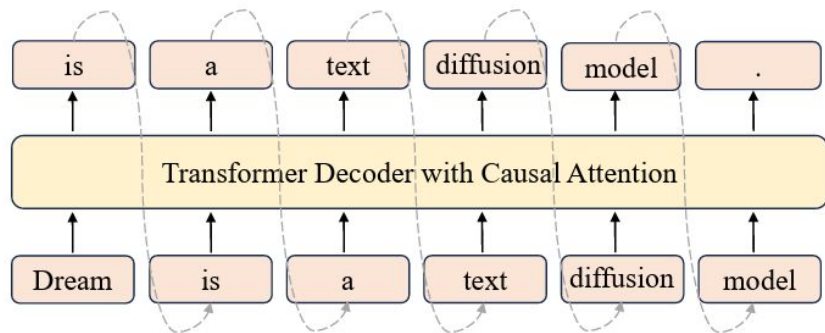
- LLaDA has impressive scalability, instruction following and in context learning capabilities
- Outperforms GPT-4o in reversal poem completion tasks
- LLaDA's sampling speed is slower than the autoregressive baseline because LLaDA samples with a fixed context length.
- LLaDA cannot yet leverage techniques like KV-Cache

	LLaDA 8B*	LLaMA3 8B*	LLaMA2 7B*	Qwen2 7B†	Qwen2.5 7B†	Mistral 7B†	Deepseek 7B*
Model	Diffusion	AR	AR	AR	AR	AR	AR
Training tokens	2.3T	15T	2T	7T	18T	-	2T
General Tasks							
MMLU	65.9 (5)	65.4 (5)	45.9 (5)	70.3 (5)	74.2 (5)	64.2 (5)	48.2 (5)
BBH	49.7 (3)	62.1 (3)	39.4 (3)	62.3 (3)	70.4 (3)	56.1 (3)	39.5 (3)
ARC-C	45.9 (0)	53.1 (0)	46.3 (0)	60.6 (25)	63.7 (25)	60.0 (25)	48.1 (0)
Hellaswag	70.5 (0)	79.1 (0)	76.0 (0)	80.7 (10)	80.2 (10)	83.3 (10)	75.4 (0)
TruthfulQA	46.1 (0)	44.0 (0)	39.0 (0)	54.2 (0)	56.4 (0)	42.2 (0)	-
WinoGrande	74.8 (5)	77.3 (5)	72.5 (5)	77.0 (5)	75.9 (5)	78.4 (5)	70.5 (0)
PIQA	73.6 (0)	80.6 (0)	79.1 (0)	-	-	-	79.2 (0)
Mathematics & Science							
GSM8K	70.3 (4)	48.7 (4)	13.1 (4)	80.2 (4)	85.4 (4)	36.2 (4)	17.4 (8)
Math	31.4 (4)	16.0 (4)	4.3 (4)	43.5 (4)	49.8 (4)	10.2 (4)	6.0 (4)
GPQA	25.2 (5)	25.9 (5)	25.7 (5)	30.8 (5)	36.4 (5)	24.7 (5)	-
Code							
HumanEval	35.4 (0)	34.8 (0)	12.8 (0)	51.2 (0)	57.9 (0)	29.3 (0)	26.2 (0)
HumanEval-FIM	73.8 (2)	73.3 (2)	26.9 (2)	-	-	-	-
MBPP	40.0 (4)	48.8 (4)	23.2 (4)	64.2 (0)	74.9 (0)	51.1 (0)	39.0 (3)
Chinese							
CMMLU	69.9 (5)	50.7 (5)	32.5 (5)	83.9 (5)	-	-	47.2 (5)
C-Eval	70.5 (5)	51.7 (5)	34.0 (5)	83.2 (5)	-	-	45.0 (5)

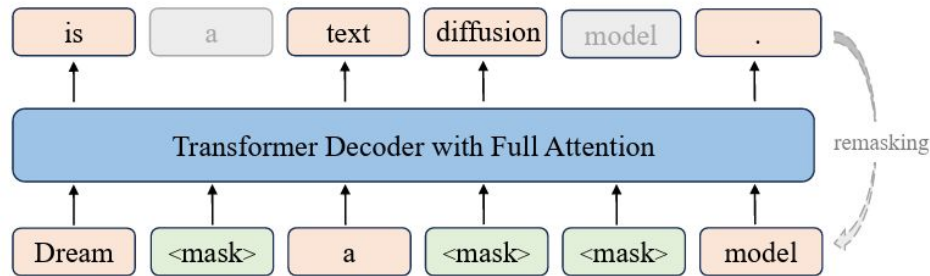


Dream-7B

- MDLM trained on 0.6T tokens instead of 2.3T
- Masked diffusion language model that uses AR-based LLM initialization instead of being trained from scratch
- Use of the hidden representation at position i to predict the token $i+1$



(a) Autoregressive Modeling



(b) Diffusion Modeling in Dream

Dream-7B

- Dream involves moving from sequence level noise scheduling to token level noise scheduling through Context Adaptive Noise Rescheduling at Token Level (CART)

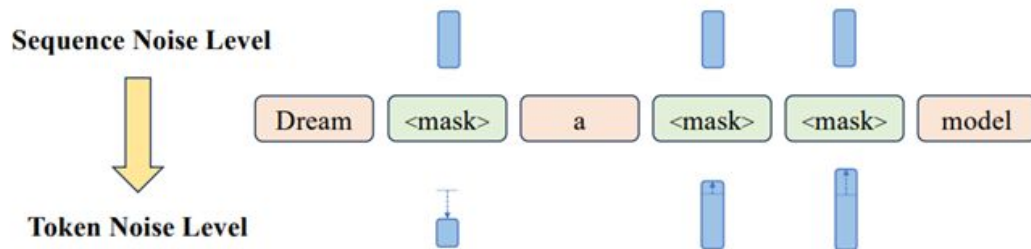


Figure 3: Illustration of the context-adaptive token-level noise rescheduling mechanism. Dream re-decides the noise level for each mask token by measuring its context informationness.

Dream 7B Evaluation

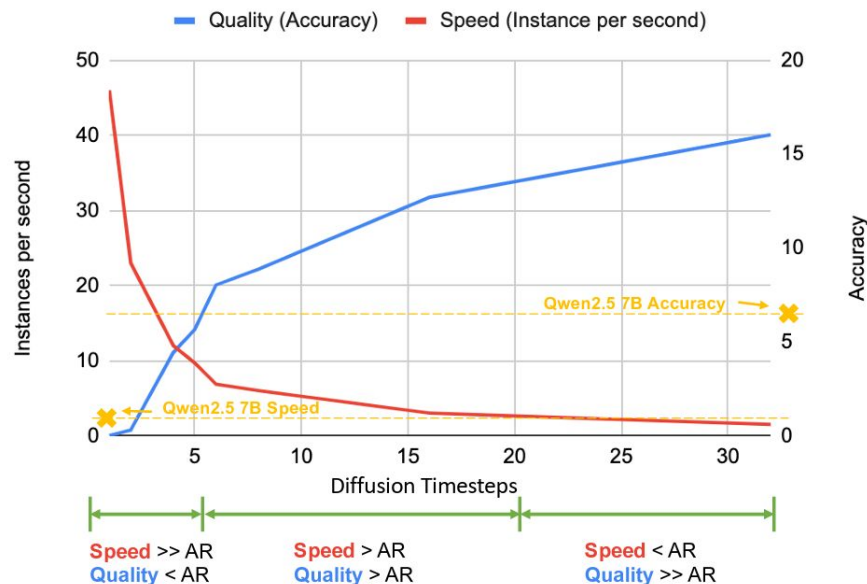
Benchmarks used:

- General Language understanding: MMLU, BBH, ARC-E, ARC-C, HellaSwag, WinoGrande, PIQA, and RACE
- Math and scientific reasoning: GSM8K, MATH, GPQA
- Code generation: HumanEval, MBPP
- Adherence to given instructions: IFEval
- Planning tasks: countdown, sudoku and trip planning

Dream 7B Results

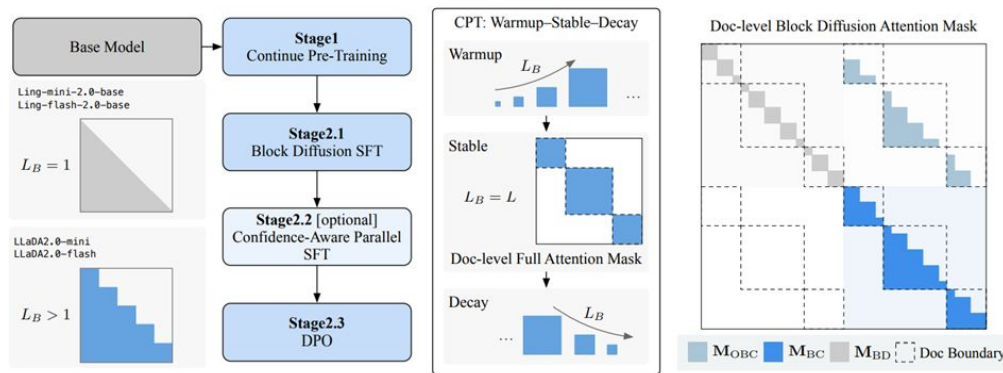
- Superior performance in reasoning and planning tasks against Qwen2.5 7B and in solving problems with multiple constraints

Model	Dream 7B*	LLaDA 8B*	Qwen2.5 7B*	LLAMA3 8B*	Mistral 7B	DeepSeek 7B
Type	Diffusion	Diffusion	AR	AR	AR	AR
Training Tokens	0.6T	2.3T	18T	15T	-	2T
General Tasks						
MMLU	69.5 (5)	65.9 (5)	71.9 (5)	63.5 (5)	60.1 (5)	48.2 (5)
BBH	57.9 (3)	47.4 (3)	63.9 (3)	62.7 (3)	-	39.5 (3)
ARC-E	83.9 (0)	71.8 (0)	77.4 (0)	81.1 (0)	80.0 (0)	67.9 (0)
ARC-C	59.8 (0)	47.5 (0)	51.5 (0)	53.6 (0)	55.5 (0)	48.1 (0)
Hellaswag	73.3 (0)	72.7 (0)	79.0 (0)	78.9 (0)	81.3 (0)	75.4 (0)
WinoGrande	74.5 (5)	73.5 (5)	76.4 (5)	76.9 (5)	75.3 (0)	70.5 (0)
PIQA	75.8 (0)	74.8 (0)	79.8 (0)	81.3 (0)	83.0 (0)	79.2 (0)
RACE	44.7 (0)	38.7 (0)	41.9 (0)	39.2 (0)	-	46.5 (5)
Mathematics & Science						
GSM8K	77.2 (8)	70.9 (8)	78.9 (8)	55.3 (8)	52.1 (8)	17.4 (8)
MATH	39.6 (4)	30.7 (4)	41.1 (4)	18.0 (4)	13.1 (4)	6.0 (4)
GPQA	36.6 (5)	30.4 (5)	35.5 (5)	30.6 (5)	-	-
Code						
HumanEval	57.9 (0)	32.9 (0)	56.7 (0)	35.4 (0)	30.5 (0)	26.2 (0)
MBPP	56.2 (4)	39.0 (4)	63.6 (4)	49.2 (4)	47.5 (3)	39.0 (3)
Planning Tasks						
Countdown	16.0 (8)	13.2 (8)	6.2 (8)	3.7 (8)	-	-
Sudoku	81.0 (8)	46.0 (8)	21.0 (8)	0.0 (8)	-	-
Trip planning	17.8 (2)	16.4 (2)	3.6 (2)	8.7 (2)	-	-



LLaDA 2.0

- Combination of MDLM and BDLM that explores AR initialized diffusion models at large scale
- Uses Warmup-Stable-Decay for the conversion from AR to dLLMs
- Uses document level attention mask to ensure semantic coherence and stability within a documents.
- Inference is done by conditioning one block on previous blocks and for each block use iterative refinement process

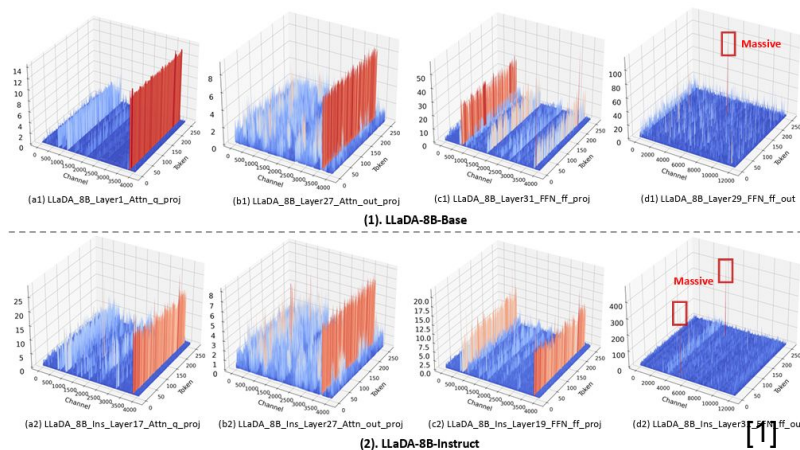


LLaDA 2.0 Results

- Evaluate the model on knowledge, reasoning, coding, math agent and alignment tasks (47 benchmarks)
- Results:
 - Llada2 mini outperforms qwen 8B in reasoning coding and math tasks
 - Llada2 flash performs better than Qwen3-30B-A3B-Instruct-2507 in coding and agents domain and complex generative tasks, but in general tasks it is still slightly worse

PTQ for Diffusion LLMs

- DLM produce normal and massive activation outliers
- Normal outliers in dLLMs have slightly lower magnitudes than in LLMs
- Massive outliers in dLLMs appear across more tokens which makes weight activation quantization more difficult as it reduce the effectiveness of scaling strategies and global clipping.



H. Lin *et al.*, “Quantization Meets dLLMs: A Systematic Study of Post-training Quantization for Diffusion LLMs,”

Oct. 15, 2025, *arXiv*: arXiv:2508.14896.

PTQ for Diffusion LLMs

- Applied weight only and weight activation techniques on DLMs.
- Benchmarks used are :
 - General knowledge tasks, such as MMLU, ARC-E, ARC-C, Hellaswag, WinoGrande, PIQA
 - Math reasoning tasks such as GSM8K, and Math
 - Code generation: HumanEval, MBPP
- Authors used the performance degradation relative to full precision models as metric for assessing quantized dLLMs. (negligible(<1%), moderate(1-4%) and significant(>4%))

PTQ for Diffusion LLMs Results

- GPTQ outperforms AWQ across most tasks.
- Rotation-based methods such as DuQuant and QuaRot demonstrate clear advantages over SmoothQuant

Table 3: Model performance on mathematics and code tasks under weight-only quantization.

Model	Setting	Method	GSM8K (4-shot) Gen Len 256	Math (0-shot) Gen Len 256	Avg	Drop	HumanEval (0-shot) Gen Len 512	MBPP (3-shot) Gen Len 512	Avg	Drop
LLaDA-8B	FP Model	-	69.7	21.3	45.5	-	32.9	39.4	36.2	-
	W4A16 g128	GPTQ	68.5	21.3	44.9	↓ 1.4%	28.7	39.4	34.0	↓ 5.9%
		AWQ	67.4	20.6	44.0	↓ 3.2%	29.9	37.2	33.5	↓ 7.3%
	W3A16 g128	GPTQ	63.3	13.4	38.4	↓ 15.7%	26.2	35.4	30.8	↓ 14.8%
		AWQ	64.3	17.0	40.6	↓ 10.7%	28.1	34.2	31.1	↓ 13.9%

Table 4: Model performance on mathematics and code tasks under weight-activation quantization.

Model	Setting	Method	GSM8K (4-shot) Gen Len 256	Math (0-shot) Gen Len 256	Avg	Drop	HumanEval (0-shot) Gen Len 512	MBPP (3-shot) Gen Len 512	Avg	Drop
LLaDA-8B	FP Model	-	69.7	21.3	45.5	-	32.9	39.4	36.2	-
	W8A8	SmoothQuant	69.4	20.2	44.8	↓ 1.6%	27.4	40.2	33.8	↓ 6.5%
		QuaRot	69.9	20.7	69.9	↓ 0.4%	31.7	40.6	36.2	↓ 0.0%
		DuQuant	70.7	20.7	45.7	↑ 0.4%	33.5	38.8	36.2	↓ 0.0%
	W4A4	SmoothQuant	0.3	2.0	1.2	↓ 97.4%	0.0	0.0	0.0	↓ 100.0%
		QuaRot	62.9	15.2	62.9	↓ 14.1%	23.8	34.6	29.2	↓ 19.3%
		DuQuant	64.4	14.8	39.6	↓ 13.0%	25.6	33.6	29.6	↓ 18.1%

Key Takeaways

- LLada models and Dream 7B are currently the best performing DLMs and are performing better than AR models in certain tasks
- Existing outlier-handling strategies may be insufficient for dLLMs in low-bit quantization regimes.
- Developing more robust 3-bit weight only and 4-bit weight activation quantization methods is an open research question
- Rotation based quantization are better in dealing with activation outliers in dLLMs.

Plan for next week:

- Focus more on quantization literature, weight only and weight activation techniques
- Study SINQ technique in depth
- Research more applications of quantization in DLMs